

Mohammadhossein Ahmadi

+1 (540) 230-9961 | ahmadimh1998@vt.edu | linkedin.com/in/mohammadhossein-ahmadi-112020103

Education

- **Virginia Tech** Blacksburg, VA
Doctor of Philosophy in Geotechnical Engineering; GPA: 3.67/4.00 Aug 2024 - Present
- **University of Tehran** Tehran, Iran
Master of Science in Geotechnical Engineering; GPA: 3.4/4.00 Sep 2021 - Aug 2024
- **Amirkabir University of Technology (Tehran Polytechnic)** Tehran, Iran
Bachelor of Science in Civil and Environmental Engineering; GPA: 3.4/4.00 Sep 2017 – Sep 2021

Experience

- **Graduate Research Assistant** Aug 2024 - Present
Virginia Tech, Blacksburg, VA Project Advisor: Dr. Joseph Vantassel
 - Problem Statement: Subsurface imaging remains a critical challenge in geotechnical engineering, particularly for accurate characterization in 2D and 3D.
 - Full Waveform Inversion (FWI) iteratively refines subsurface models by minimizing differences between observed and simulated waveforms, improving the resolution of geological features and material properties.
 - FWI supports enhanced imaging that accurately delineates subsurface anomalies and structures and informs design decisions and risk assessments.
- **Graduate Research Assistant** Jan 2022 - Aug 2024
University of Tehran, Tehran, Iran Project Advisor: Dr. Reza Rafiee
 - Implementation of various formulations for modeling infinite boundaries in direct SSI methods using 2.5-dimensional finite element models in porous saturated half-space media
- **Undergraduate project** Sep 2020 – Sep 2021
Amirkabir University of Technology, Tehran, Iran Project Advisor: Siamak Epackachi
 - Developing design and analysis online application for Bolted Unstiffened and Stiffened Extended End-Plate Moment Connections

Professional Experience

- **Application Developer in CIVision Civil Department** Nov2020 - Jul2022
CIVision Group (online design & analysis application for connections), Tehran, Iran

Technical Skills

- **Software and Tools:** COMSOL Multiphysics, ABAQUS, GeoStudio, Neural Networks, Machine Learning
- **Languages:** Python, MATLAB
- **Relevant Courses:** Advanced Soil Mechanics, Continuum Mechanics and Soil Plasticity, Application of Numerical Methods in Geotechnics